Frank-Wolfe

The algorithm

In terms of computation, FW replaces Projection with linear minimization.

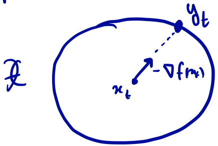
$$\min: f(x) \qquad f \text{ is } \beta\text{-smooth} \quad (*)$$
$$\text{st}: x \in \mathcal{X}$$

Projected GD: $x_{t+1} = \mathrm{Proj}_{\mathcal{X}}(x_t - \eta \nabla f(x_t))$

FW: $y_t \in \operatorname{argmin}: \nabla f(x_t)^T y$
$\qquad \text{st}: y \in \mathcal{X}$

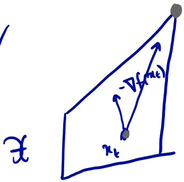
$$x_{t+1} = (1-\gamma_t) x_t + \gamma_t \cdot y_t$$

① $\mathcal{X}$ = sphere around origin.



looks very similar to GD.

$y_t = \operatorname{argmin}: \nabla f(x_t)^T y$



FW in this example does not follow the gradient trajectory

The main result

Thm: Let $f, \mathcal{X}$ be convex, with f β-smooth in any arbitrary norm : $\|\nabla f(x) - \nabla f(y)\|_* \le \beta \cdot \|x-y\|$

min: $f(x)$
st: $x \in \mathcal{X}$

$R = \operatorname{diam} \mathcal{X} = \sup_{x,y \in \mathcal{X}} \|x-y\|$

$\gamma_t = \frac{2}{t+1}$.

Then, the FW alg : $\boxed{y_t \in \operatorname{argmin}_{x \in \mathcal{X}} : \nabla f(x_t)^T y ,}$

$\longrightarrow \quad x_{t+1} = (1-\gamma_t) x_t + \gamma_t y_t$

guarantees : $\boxed{f(x_T) - f(x^*) \le \frac{2 \beta R^2}{T+1}}$

① $O(1/T)$
$\le$
$O(1/\varepsilon)$

② β is smoothness in any norm of choice.

Note: smoothness in any norm

Frank-Wolfe proof 1/2

Recall: If f is β-smooth, i.e., $\|\nabla f(x) - \nabla f(y)\|_* \leq \beta \cdot \|x-y\|$

Then we have a quadratic upper bound:

$$f(y) \leq f(x) + \nabla f(x)^T (y-x) + \frac{\beta}{2} \|y-x\|^2$$

Applying this:

$$f(x_{t+1}) - f(x_t) \leq \nabla f(x_t)^T (x_{t+1} - x_t) + \frac{\beta}{2} \|x_{t+1} - x_t\|^2$$

used def of FW $\longrightarrow$ $\leq \gamma_t \cdot \underbrace{\nabla f(x_t)^T (y_t - x_t)}_{\nabla f(x_t)^T y_t} + \frac{\beta}{2} \gamma_t^2 \cdot R^2$ $\quad (x_{t+1} = (1-\gamma_t)x_t + \gamma_t \cdot y_t)$

also comes from def of FW $\longrightarrow$ $\leq \gamma_t \cdot \underbrace{\nabla f(x_t)^T (x^* - x_t)}_{\nabla f(x_t)^T x^*} + \frac{\beta}{2} \gamma_t^2 \cdot R^2$

$$\leq \gamma_t \left(f(x^*) - f(x_t) \right) + \frac{\beta}{2} \gamma_t^2 R^2$$

$$\Rightarrow \underbrace{f(x_{t+1}) - f(x^*) \leq (1-\gamma_t)\left(f(x_t) - f(x^*) \right) + \frac{\beta}{2} \gamma_t^2 R^2}$$

Frank-Wolfe proof 2/2

So far: $\underbrace{f(x_{t+1}) - f(x^*)}_{\delta_{t+1}} \leq (1-\gamma_t)\underbrace{(f(x_t) - f(x^*))}_{\delta_t} + \frac{\beta}{2}\gamma_t^2 R^2$. (*)

We want to show: $\delta_T \leq \frac{2\beta R^2}{T+1}$

proof by a simple induction.

Check: true for $t=2$. i.e., $\delta_2 \leq \frac{2\beta R^2}{3}$

Inductive hypothesis: assume true for all t

Need to show it holds for $t+1$.

$$\delta_{t+1} \leq (1-\gamma_t)\,\delta_t + \frac{\beta}{2}\gamma_t^2 R^2 .$$

recall: $\gamma_t = \frac{2}{t+1}$

$$\leq \left(1 - \frac{2}{t+1}\right)\frac{2\beta R^2}{t+1} + \frac{\beta}{2}\left(\frac{2}{t+1}\right)^2 R^2$$

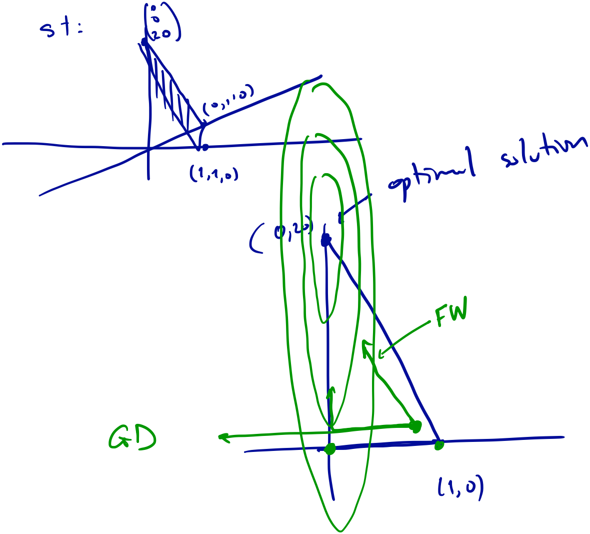
$$= \frac{t-1}{t+1}\cdot\frac{2\beta R^2}{t+1} + \frac{2\beta R^2}{(t+1)^2} = \frac{(t-1+1)\cdot 2\beta R^2}{(t+1)^2} = \frac{2\beta R^2}{t+1}\cdot\frac{t}{t+1}$$

$$\leq \frac{2\beta R^2}{t+1}.$$

Frank-Wolfe Example 1/2

min: $f(x_1, x_2, x_3) = 100 \cdot x_1^2 + x_2^2 + (x_3 - 20)^2$

st:



Frank-Wolfe Example 2/2

GD

FW